

task-specific [SDCW19, JYS⁺19, KR16] approaches to distillation of language models. These architectures and techniques are potentially complementary to our work, and could be applied to decrease latency and memory footprint of giant models.

As fine-tuned language models have neared human performance on many standard benchmark tasks, considerable effort has been devoted to constructing more difficult or open-ended tasks, including question answering [KPR⁺19, IBGC⁺14, CCE⁺18, MCKS18], reading comprehension [CHI⁺18, RCM19], and adversarially constructed datasets designed to be difficult for existing language models [SBBC19, NWD⁺19]. In this work we test our models on many of these datasets.

Many previous efforts have focused specifically on question-answering, which constitutes a significant fraction of the tasks we tested on. Recent efforts include [RSR⁺19, RRS20], which fine-tuned an 11 billion parameter language model, and [GLT⁺20], which focused on attending over a large corpus of data at test time. Our work differs in focusing on in-context learning but could be combined in the future with those of [GLT⁺20, LPP⁺20].

Metalearning in language models has been utilized in [RWC⁺19], though with much more limited results and no systematic study. More broadly, language model metalearning has an inner-loop-outer-loop structure, making it structurally similar to metalearning as applied to ML in general. Here there is an extensive literature, including matching networks [VBL⁺16], RL2 [DSC⁺16], learning to optimize [RL16, ADG⁺16, LM17] and MAML [FAL17]. Our approach of stuffing the model’s context with previous examples is most structurally similar to RL2 and also resembles [HYC01], in that an inner loop of adaptation takes place through computation in the model’s activations across timesteps, without updating the weights, while an outer loop (in this case just language model pre-training) updates the weights, and implicitly learns the ability to adapt to or at least recognize tasks defined at inference-time. Few-shot auto-regressive density estimation was explored in [RCP⁺17] and [GWC⁺18] studied low-resource NMT as a few-shot learning problem.

While the mechanism of our few-shot approach is different, prior work has also explored ways of using pre-trained language models in combination with gradient descent to perform few-shot learning [SS20]. Another sub-field with similar goals is semi-supervised learning where approaches such as UDA [XDH⁺19] also explore methods of fine-tuning when very little labeled data is available.

Giving multi-task models instructions in natural language was first formalized in a supervised setting with [MKXS18] and utilized for some tasks (such as summarizing) in a language model with [RWC⁺19]. The notion of presenting tasks in natural language was also explored in the text-to-text transformer [RSR⁺19], although there it was applied for multi-task fine-tuning rather than for in-context learning without weight updates.

Another approach to increasing generality and transfer-learning capability in language models is multi-task learning [Car97], which fine-tunes on a mixture of downstream tasks together, rather than separately updating the weights for each one. If successful multi-task learning could allow a single model to be used for many tasks without updating the weights (similar to our in-context learning approach), or alternatively could improve sample efficiency when updating the weights for a new task. Multi-task learning has shown some promising initial results [LGH⁺15, LSP⁺18] and multi-stage fine-tuning has recently become a standardized part of SOTA results on some datasets [PFB18] and pushed the boundaries on certain tasks [KKS⁺20], but is still limited by the need to manually curate collections of datasets and set up training curricula. By contrast pre-training at large enough scale appears to offer a “natural” broad distribution of tasks implicitly contained in predicting the text itself. One direction for future work might be attempting to generate a broader set of explicit tasks for multi-task learning, for example through procedural generation [TFR⁺17], human interaction [ZSW⁺19b], or active learning [Mac92].

Algorithmic innovation in language models over the last two years has been enormous, including denoising-based bidirectionality [DCLT18], prefixLM [DL15] and encoder-decoder architectures [LLG⁺19, RSR⁺19], random permutations during training [YDY⁺19], architectures that improve the efficiency of sampling [DYY⁺19], improvements in data and training procedures [LOG⁺19], and efficiency increases in the embedding parameters [LCG⁺19]. Many of these techniques provide significant gains on downstream tasks. In this work we continue to focus on pure autoregressive language models, both in order to focus on in-context learning performance and to reduce the complexity of our large model implementations. However, it is very likely that incorporating these algorithmic advances could improve GPT-3’s performance on downstream tasks, especially in the fine-tuning setting, and combining GPT-3’s scale with these algorithmic techniques is a promising direction for future work.

8 Conclusion

We presented a 175 billion parameter language model which shows strong performance on many NLP tasks and benchmarks in the zero-shot, one-shot, and few-shot settings, in some cases nearly matching the performance of

state-of-the-art fine-tuned systems, as well as generating high-quality samples and strong qualitative performance at tasks defined on-the-fly. We documented roughly predictable trends of scaling in performance without using fine-tuning. We also discussed the social impacts of this class of model. Despite many limitations and weaknesses, these results suggest that very large language models may be an important ingredient in the development of adaptable, general language systems.

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Contributions

Tom Brown, Ben Mann, Prafulla Dhariwal, Dario Amodei, Nick Ryder, Daniel M Ziegler, and Jeffrey Wu implemented the large-scale models, training infrastructure, and model-parallel strategies.

Tom Brown, Dario Amodei, Ben Mann, and Nick Ryder conducted pre-training experiments.

Ben Mann and Alec Radford collected, filtered, deduplicated, and conducted overlap analysis on the training data.

Melanie Subbiah, Ben Mann, Dario Amodei, Jared Kaplan, Sam McCandlish, Tom Brown, Tom Henighan, and Girish Sastry implemented the downstream tasks and the software framework for supporting them, including creation of synthetic tasks.

Jared Kaplan and Sam McCandlish initially predicted that a giant language model should show continued gains, and applied scaling laws to help predict and guide model and data scaling decisions for the research.

Ben Mann implemented sampling without replacement during training.

Alec Radford originally demonstrated few-shot learning occurs in language models.

Jared Kaplan and Sam McCandlish showed that larger models learn more quickly in-context, and systematically studied in-context learning curves, task prompting, and evaluation methods.

Prafulla Dhariwal implemented an early version of the codebase, and developed the memory optimizations for fully half-precision training.

Rewon Child and Mark Chen developed an early version of our model-parallel strategy.

Rewon Child and Scott Gray contributed the sparse transformer.

Aditya Ramesh experimented with loss scaling strategies for pretraining.

Melanie Subbiah and Arvind Neelakantan implemented, experimented with, and tested beam search.

Pranav Shyam worked on SuperGLUE and assisted with connections to few-shot learning and meta-learning literature.

Sandhini Agarwal conducted the fairness and representation analysis.

Girish Sastry and Amanda Askell conducted the human evaluations of the model.

Ariel Herbert-Voss conducted the threat analysis of malicious use.

Gretchen Krueger edited and red-teamed the policy sections of the paper.

Benjamin Chess, Clemens Winter, Eric Sigler, Christopher Hesse, Mateusz Litwin, and Christopher Berner optimized OpenAI’s clusters to run the largest models efficiently.

Scott Gray developed fast GPU kernels used during training.

Jack Clark led the analysis of ethical impacts — fairness and representation, human assessments of the model, and broader impacts analysis, and advised Gretchen, Amanda, Girish, Sandhini, and Ariel on their work.

Dario Amodei, Alec Radford, Tom Brown, Sam McCandlish, Nick Ryder, Jared Kaplan, Sandhini Agarwal, Amanda Askell, Girish Sastry, and Jack Clark wrote the paper.

Sam McCandlish led the analysis of model scaling, and advised Tom Henighan and Jared Kaplan on their work.

Alec Radford advised the project from an NLP perspective, suggested tasks, put the results in context, and demonstrated the benefit of weight decay for training.

Ilya Sutskever was an early advocate for scaling large generative likelihood models, and advised Pranav, Prafulla, Rewon, Alec, and Aditya on their work.

Dario Amodei designed and led the research.

A Details of Common Crawl Filtering

As mentioned in Section 2.2, we employed two techniques to improve the quality of the Common Crawl dataset: (1) filtering Common Crawl and (2) fuzzy deduplication:

1. In order to improve the quality of Common Crawl, we developed an automatic filtering method to remove low quality documents. Using the original WebText as a proxy for high-quality documents, we trained a classifier to distinguish these from raw Common Crawl. We then used this classifier to re-sample Common Crawl by prioritizing documents which were predicted by the classifier to be higher quality. The classifier is trained using logistic regression classifier with features from Spark’s standard tokenizer and HashingTF¹⁰. For the positive examples, we used a collection of curated datasets such as WebText, Wikiedia, and our web books corpus as the positive examples, and for the negative examples, we used unfiltered Common Crawl. We used this classifier to score Common Crawl documents. We kept each document in our dataset iff

$$\text{np.random.pareto}(\alpha) > 1 - \text{document_score}$$

We chose $\alpha = 9$ in order to take mostly documents the classifier scored highly, but still include some documents that were out of distribution. α was chosen to match the distribution of scores from our classifier on WebText. We found this re-weighting increased quality as measured by loss on a range of out-of-distribution generative text samples.

2. To further improve model quality and prevent overfitting (which becomes increasingly important as model capacity increases), we fuzzily deduplicated documents (i.e. removed documents with high overlap with other documents) within each dataset using Spark’s MinHashLSH implementation with 10 hashes, using the same features as were used for classification above. We also fuzzily removed WebText from Common Crawl. Overall this decreased dataset size by an average of 10%.

After filtering for duplicates and quality, we also partially removed text occurring in benchmark datasets, described in Appendix C.

B Details of Model Training

To train all versions of GPT-3, we use Adam with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and $\epsilon = 10^{-8}$, we clip the global norm of the gradient at 1.0, and we use cosine decay for learning rate down to 10% of its value, over 260 billion tokens (after 260 billion tokens, training continues at 10% of the original learning rate). There is a linear LR warmup over the first 375 million tokens. We also gradually increase the batch size linearly from a small value (32k tokens) to the full value over the first 4-12 billion tokens of training, depending on the model size. Data are sampled without replacement during training (until an epoch boundary is reached) to minimize overfitting. All models use weight decay of 0.1 to provide a small amount of regularization [LH17].

During training we always train on sequences of the full $n_{\text{ctx}} = 2048$ token context window, packing multiple documents into a single sequence when documents are shorter than 2048, in order to increase computational efficiency. Sequences with multiple documents are not masked in any special way but instead documents within a sequence are delimited with a special end of text token, giving the language model the information necessary to infer that context separated by the end of text token is unrelated. This allows for efficient training without need for any special sequence-specific masking.

C Details of Test Set Contamination Studies

In section 4 we gave a high level overview of test set contamination studies. In this section we provide details on methodology and results.

Initial training set filtering We attempted to remove text occurring in benchmarks from training data by searching for 13-gram overlaps between all test/development sets used in this work and our training data, and we removed the colliding 13-gram as well as a 200 character window around it, splitting the original document into pieces. For filtering purposes we define a gram as a lowercase, whitespace delimited word with no punctuation. Pieces less than 200 characters long were discarded. Documents split into more than 10 pieces were considered contaminated and

¹⁰<https://spark.apache.org/docs/latest/api/python/pyspark.ml.html#pyspark.ml.feature.HashingTF>

removed entirely. Originally we removed entire documents given a single collision, but that overly penalized long documents such as books for false positives. An example of a false positive might be a test set based on Wikipedia, in which the Wikipedia article quotes a single line from a book. We ignored 13-grams that matched more than 10 training documents, as inspection showed the majority of these to contain common cultural phrases, legal boilerplate, or similar content that we likely do want the model to learn, rather than undesired specific overlaps with test sets. Examples for various frequencies can be found in the GPT-3 release repository¹¹.

Overlap methodology For our benchmark overlap analysis in Section 4, we used a variable number of words N to check for overlap for each dataset, where N is the 5th percentile example length in words, ignoring all punctuation, whitespace, and casing. Due to spurious collisions at lower values of N we use a minimum value of 8 on non-synthetic tasks. For performance reasons, we set a maximum value of 13 for all tasks. Values for N and the amount of data marked as dirty are shown in Table C.1. Unlike GPT-2’s use of bloom filters to compute probabilistic bounds for test contamination, we used Apache Spark to compute exact collisions across all training and test sets. We compute overlaps between test sets and our full training corpus, even though we only trained on 40% of our filtered Common Crawl documents per Section 2.2.

We define a ‘dirty’ example as one with any N -gram overlap with any training document, and a ‘clean’ example as one with no collision.

Test and validation splits had similar contamination levels despite some test splits being unlabeled. Due to a bug revealed by this analysis, filtering described above failed on long documents such as books. Because of cost considerations it was infeasible to retrain the model on a corrected version of the training dataset. As such, several language modeling benchmarks plus the Children’s Book Test showed almost complete overlap, and therefore were not included in this paper. Overlaps are shown in Table C.1

Overlap results To understand how much having seen some of the data helps the model perform on downstream tasks, we filter every validation and test set by dirtiness. Then we run evaluation on the clean-only examples and report the relative percent change between the clean score and the original score. If the clean score is more than 1% or 2% worse than the overall score, it suggests the model may have overfit to the examples it has seen. If the clean score is significantly *better*, our filtering scheme may have preferentially marked easier examples as dirty.

This overlap metric tends to show a high rate of false positives for datasets that contain background information (but not answers) drawn from the web (such as SQuAD, which draws from Wikipedia) or examples less than 8 words long, which we ignored in our filtering process (except for wordscrambling tasks). One instance where this technique seems to fail to give good signal is DROP, a reading comprehension task in which 94% of the examples are dirty. The information required to answer the question is in a passage provided to the model, so having seen the passage during training but not the questions and answers does not meaningfully constitute cheating. We confirmed that every matching training document contained only the source passage, and none of the questions and answers in the dataset. The more likely explanation for the decrease in performance is that the 6% of examples that remain after filtering come from a slightly different distribution than the dirty examples.

Figure 4.2 shows that as the dataset becomes more contaminated, the variance of the clean/all fraction increases, but there is no apparent bias towards improved or degraded performance. This suggests that GPT-3 is relatively insensitive to contamination. See Section 4 for details on the datasets we flagged for further review.

¹¹https://github.com/openai/gpt-3/blob/master/overlap_frequency.md

Name	Split	Metric	<i>N</i>	Acc/F1/BLEU	Total Count	Dirty Acc/F1/BLEU	Dirty Count	Clean Acc/F1/BLEU	Clean Count	Clean Percentage	Relative Difference Clean vs All
Quac	dev	f1	13	44.3	7353	44.3	7315	54.1	38	1%	20%
SQuADv2	dev	f1	13	69.8	11873	69.9	11136	68.4	737	6%	-2%
DROP	dev	f1	13	36.5	9536	37.0	8898	29.5	638	7%	-21%
Symbol Insertion	dev	acc	7	66.9	10000	66.8	8565	67.1	1435	14%	0%
CoQa	dev	f1	13	86.0	7983	85.3	5107	87.1	2876	36%	1%
ReCoRD	dev	acc	13	89.5	10000	90.3	6110	88.2	3890	39%	-1%
Winograd	test	acc	9	88.6	273	90.2	164	86.2	109	40%	-3%
BoolQ	dev	acc	13	76.0	3270	75.8	1955	76.3	1315	40%	0%
MultiRC	dev	acc	13	74.2	953	73.4	558	75.3	395	41%	1%
RACE-h	test	acc	13	46.8	3498	47.0	1580	46.7	1918	55%	0%
LAMBADA	test	acc	13	86.4	5153	86.9	2209	86.0	2944	57%	0%
LAMBADA (No Blanks)	test	acc	13	77.8	5153	78.5	2209	77.2	2944	57%	-1%
WSC	dev	acc	13	76.9	104	73.8	42	79.0	62	60%	3%
PIQA	dev	acc	8	82.3	1838	89.9	526	79.3	1312	71%	-4%
RACE-m	test	acc	13	58.5	1436	53.0	366	60.4	1070	75%	3%
De→En 16	test	bleu-sb	12	43.0	2999	47.4	739	40.8	2260	75%	-5%
En→De 16	test	bleu-sb	12	30.9	2999	32.6	739	29.9	2260	75%	-3%
En→Ro 16	test	bleu-sb	12	25.8	1999	24.9	423	26.1	1576	79%	1%
Ro→En 16	test	bleu-sb	12	41.3	1999	40.4	423	41.6	1576	79%	1%
WebQs	test	acc	8	41.5	2032	41.6	428	41.5	1604	79%	0%
ANLI R1	test	acc	13	36.8	1000	40.5	200	35.9	800	80%	-3%
ANLI R2	test	acc	13	34.0	1000	29.4	177	35.0	823	82%	3%
TriviaQA	dev	acc	10	71.2	7993	70.8	1390	71.3	6603	83%	0%
ANLI R3	test	acc	13	40.2	1200	38.3	196	40.5	1004	84%	1%
En→Fr 14	test	bleu-sb	13	39.9	3003	38.3	411	40.3	2592	86%	1%
Fr→En 14	test	bleu-sb	13	41.4	3003	40.9	411	41.4	2592	86%	0%
WiC	dev	acc	13	51.4	638	53.1	49	51.3	589	92%	0%
RTE	dev	acc	13	71.5	277	71.4	21	71.5	256	92%	0%
CB	dev	acc	13	80.4	56	100.0	4	78.8	52	93%	-2%
Anagrams 2	dev	acc	2	40.2	10000	76.2	705	37.4	9295	93%	-7%
Reversed Words	dev	acc	2	0.4	10000	1.5	660	0.3	9340	93%	-26%
OpenBookQA	test	acc	8	65.4	500	58.1	31	65.9	469	94%	1%
ARC (Easy)	test	acc	11	70.1	2268	77.5	89	69.8	2179	96%	0%
Anagrams 1	dev	acc	2	15.0	10000	49.8	327	13.8	9673	97%	-8%
COPA	dev	acc	9	93.0	100	100.0	3	92.8	97	97%	0%
ARC (Challenge)	test	acc	12	51.6	1144	45.2	31	51.8	1113	97%	0%
HellaSwag	dev	acc	13	79.3	10042	86.2	152	79.2	9890	98%	0%
NQs	test	acc	11	29.9	3610	32.7	52	29.8	3558	99%	0%
Cycled Letters	dev	acc	2	38.6	10000	20.5	73	38.7	9927	99%	0%
SAT Analogies	dev	acc	9	65.8	374	100.0	2	65.6	372	99%	0%
StoryCloze	test	acc	13	87.7	1871	100.0	2	87.6	1869	100%	0%
Winogrande	dev	acc	13	77.7	1267	-	0	77.7	1267	100%	0%

Table C.1: Overlap statistics for all datasets sorted from dirtiest to cleanest. We consider a dataset example dirty if it has a single *N*-gram collision with any document in our training corpus. “Relative Difference Clean vs All” shows the percent change in performance between only the clean examples vs all the examples in the benchmark. “Count” shows the number of examples. “Clean percentage” is the percent of examples that are clean vs total. For “Acc/F1/BLEU” we use the metric specified in “Metric”. These scores come from evaluations with a different seed for the random examples used for in-context learning, and will therefore differ slightly from the scores elsewhere in the paper.

D Total Compute Used to Train Language Models

This appendix contains the calculations that were used to derive the approximate compute used to train the language models in Figure 2.2. As a simplifying assumption, we ignore the attention operation, as it typically uses less than 10% of the total compute for the models we are analyzing.

Calculations can be seen in Table D.1 and are explained within the table caption.

Model	Total train compute (PF-days)	Total train compute (flops)	Params (M)	Training tokens (billions)	Flops per param per token	Mult for bwd pass	Fwd-pass flops per active param per token	Frac of params active for each token
T5-Small	2.08E+00	1.80E+20	60	1,000	3	3	1	0.5
T5-Base	7.64E+00	6.60E+20	220	1,000	3	3	1	0.5
T5-Large	2.67E+01	2.31E+21	770	1,000	3	3	1	0.5
T5-3B	1.04E+02	9.00E+21	3,000	1,000	3	3	1	0.5
T5-11B	3.82E+02	3.30E+22	11,000	1,000	3	3	1	0.5
BERT-Base	1.89E+00	1.64E+20	109	250	6	3	2	1.0
BERT-Large	6.16E+00	5.33E+20	355	250	6	3	2	1.0
RoBERTa-Base	1.74E+01	1.50E+21	125	2,000	6	3	2	1.0
RoBERTa-Large	4.93E+01	4.26E+21	355	2,000	6	3	2	1.0
GPT-3 Small	2.60E+00	2.25E+20	125	300	6	3	2	1.0
GPT-3 Medium	7.42E+00	6.41E+20	356	300	6	3	2	1.0
GPT-3 Large	1.58E+01	1.37E+21	760	300	6	3	2	1.0
GPT-3 XL	2.75E+01	2.38E+21	1,320	300	6	3	2	1.0
GPT-3 2.7B	5.52E+01	4.77E+21	2,650	300	6	3	2	1.0
GPT-3 6.7B	1.39E+02	1.20E+22	6,660	300	6	3	2	1.0
GPT-3 13B	2.68E+02	2.31E+22	12,850	300	6	3	2	1.0
GPT-3 175B	3.64E+03	3.14E+23	174,600	300	6	3	2	1.0

Table D.1: Starting from the right hand side and moving left, we begin with the number of training tokens that each model was trained with. Next we note that since T5 uses an encoder-decoder model, only half of the parameters are active for each token during a forward or backwards pass. We then note that each token is involved in a single addition and a single multiply for each active parameter in the forward pass (ignoring attention). Then we add a multiplier of 3x to account for the backwards pass (as computing both $\frac{\partial \text{params}}{\partial \text{loss}}$ and $\frac{\partial \text{acts}}{\partial \text{loss}}$ use a similar amount of compute as the forwards pass. Combining the previous two numbers, we get the total flops per parameter per token. We multiply this value by the total training tokens and the total parameters to yield the number of total flops used during training. We report both flops and petaflop/s-day (each of which are $8.64\text{e}+19$ flops).

E Human Quality Assessment of Synthetic News Articles

This appendix contains details on the experiments measuring human ability to distinguish GPT-3-generated synthetic news articles from real news articles. We first describe the experiments on the ~ 200 word news articles, and then describe the preliminary investigation of ~ 500 word news articles generated by GPT-3.

Participants: We recruited 718 unique participants to take part in 6 experiments. 97 participants were excluded for failing an internet check question, leaving a total of 621 participants: 343 male, 271 female, and 7 other. Mean participant age was ~ 38 years old. All participants were recruited through Positly, which maintains a whitelist of high-performing workers from Mechanical Turk. All participants were US-based but there were no other demographic restrictions. Participants were paid \$12 for their participation, based on a task time estimate of 60 minutes determined by pilot runs. In order to ensure that the sample of participants for each experiment quiz was unique, participants were not allowed to take part in an experiment more than once.

Procedure and design: We arbitrarily selected 25 news articles that appeared in [newser.com](https://www.newser.com) in early 2020. We used the article titles and subtitles to produce outputs from the 125M, 350M, 760M, 1.3B, 2.7B, 6.7B, 13.0B, and 200B (GPT-3) parameter language models. Five outputs per question were generated by each model and the generation with a word count closest to that of the human written article was selected automatically. This was to minimize the effect that completion length might have on participants' judgments. The same output procedure for each model with the exception of the removal of the intentionally bad control model, as described in the main text.

Model	Participants Recruited	Participants Excluded	Genders (m:f:other)	Mean Age	Average Word Count (human:model)
Control	76	7	32:37:0	39	216:216
GPT-3 Small	80	7	41:31:1	40	216:188
GPT-3 Medium	80	7	46:28:2	39	216:202
GPT-3 Large	81	24	46:28:2	37	216:200
GPT-3 XL	79	14	32:32:1	38	216:199
GPT-3 2.7B	80	11	36:33:0	40	216:202
GPT-3 6.7B	76	5	46:28:2	37	216:195
GPT-3 13.0B	81	13	46:28:2	37	216:209
GPT-3 175B	80	9	42:29:0	37	216:216

Table E.1: Participant details and article lengths for each experiment to evaluate human detection of ~ 200 word model generated news articles. Participants were excluded due to internet check fails.

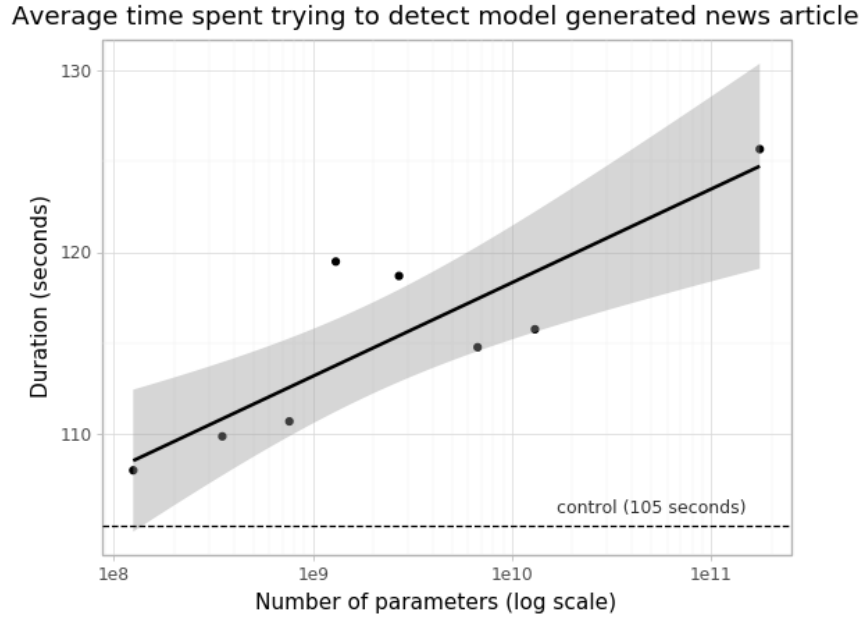


Figure E.1: Participants spend more time trying to identify whether each news article is machine generated as model size increases. Duration on the control model is indicated with the dashed line. Line of best fit is a linear model on a log scale with 95% confidence intervals.

In each experiment, half of the participants were randomly assigned to quiz A and half were randomly assigned to quiz B. Each quiz consisted of 25 articles: half (12-13) were human written and half (12-13) were model generated: the articles with human written completions in quiz A had model generated completions in quiz B and vice versa. The order of quiz question was shuffled for each participant. Participants could leave comments and were asked to indicate if they had seen the articles before. Participants were instructed not to look up the articles or their content during the quiz and at the end of the quiz were asked if they had looked anything up during the quiz.

Statistical Tests: To compare means on the different runs, we performed a two-sample t-test for independent groups for each model against the control. This was implemented in Python using the `scipy.stats.ttest_ind` function. When plotting a regression line in the graph of average participant accuracy vs model size, we fit a power law of the form ax^{-b} . The 95% confidence intervals were estimated from the t-distribution of the sample mean.

Duration statistics: In the main text, we discussed the finding that the ability of human participants to distinguish model and human generated news articles decreases as our models become larger. We have also found that the average time spent for a given set of questions increases as the model size increases, as shown in Figure E.1. Lower

Model	Participants Recruited	Participants Excluded	Genders (m:f:other)	Mean Age	Average Word Count (human:model)
Control	79	17	32:37:0	39	569:464
GPT-3 175B	81	19	32:30:0	40	569:498

Table E.2: Participant details and article lengths for the experiments investigating human detection of ~ 500 word model generated news articles. Participants were excluded due to internet check fails.

accuracy scores despite increased time investment from participants supports the finding that larger models generate harder-to-distinguish news articles.

Preliminary investigation of ~ 500 word articles: We recruited 160 unique US-based participants to take part in 2 experiments through Positly (details are given in Table E.2). We randomly selected 12 Reuters world news articles from late 2019 and created a context for GPT-3 175B that consisted of a single Reuters article not in this set of 12. We then used the article titles and Reuters locations to generate completions from GPT-3 175B and the 160M control model from the previous experiments. These were used to create two 12-question quizzes per model, each consisting of half human written and half model generated articles. Comprehension questions were added and articles were shown to participants in 3 stages at 30 second intervals to encourage closer reading. Participants were paid \$12 for this task. Model generation selection methods, exclusion criteria, and statistical tests mirror those of the previous experiments.

F Additional Samples from GPT-3

GPT-3 adapts well to many tasks other than the ones explored in the main body of the paper. As an example, in Figure F.1, we show four uncensored samples from a prompt suggesting that the model write a poem, with a given title, in the style of Wallace Stevens. We first experimented with a few prompts, then generated four samples with no additional editing or selection (sampling at temperature 1 using nucleus sampling [HBFC19] with $P = 0.9$). Completions were truncated when the model began to write a new title and author heading, or broke into prose commentary.

Context →	The City BY C. P. CAVAFY TRANSLATED BY EDMUND KEELEY [Poem text omitted] SOME TREES John Ashbery [Poem text omitted] Shadows on the Way Wallace Stevens
----- Generated Poem 1 -----	----- Generated Poem 3 -----
I must have shadows on the way If I am to walk I must have Each step taken slowly and alone To have it ready made And I must think in lines of grey To have dim thoughts to be my guide Must look on blue and green And never let my eye forget That color is my friend And purple must surround me too The yellow of the sun is no more Intrusive than the bluish snow That falls on all of us. I must have Grey thoughts and blue thoughts walk with me If I am to go away at all.	The sun was all we had. Now, in the shade All is changed. The mind must dwell on those White fields, that to its eyes were always old; Those ancient gleams, convoluting The way they lay among the huge roots, The great dark tomes of reverie, The plumed labyrinth of the sea. When our own breath gone faint and fade Up the long stairway of the dead We shall remember well The blindness of the light, the whiteness Of that white land. The footsteps, and The grasses in that light, and in the shade The menacing life. We played, too, once, in the warmest rooms. To one content, there is one content For all mankind. The forest glades Are all the more for being in fable, And wood and town in legend shadowed- Once more to return. The one content Has in it every happiness, more brave For being purposeless; and so Those island gardens, deep in the mind, Make of remoteness familiarity And of the unspoken name a friend.
----- Generated Poem 2 -----	----- Generated Poem 4 -----
There is a way in the world. That way a man may live in a small town His lifetime and see no man come, Never the stir of men about a market, Never the bobbing of covered carts, Never the creak passed on the stair Closing some other life; an old man Can prosper in that town, can prosper When his store is seen to close, his wife Fades in his house, and the world shouts in vain That no one comes to hear him Where his small bell tolls. In that small town Everything is known. The ways of the young men are known, And the ways of the young women As they walk to the Spring; The rumors are all known, The quarrels, the hurt, the envy Of the men and the women. Till a day comes That way out of the world That leads to a wood. Young, the young man Walks, a lover now, Knows the streets as they are and is of them. Comes to the wood. There, in the wood, among trees, He sees shadows on the way, hears voices, hears the wind and the rustling of leaves; Through an open glade He sees a shape and the shape hears: It waits as he waits, Waits as the shadows wait, As the voices wait; Shadows on the way, voices in the wind.	Nobody will come to this place. It is a road that leads nowhere. The solitude is deep. The mountains are high. But they are desolate, and they turn the traveler's face Towards the North. All the sounds of the world are far away. When the wind rises above the trees, The boughs bow to the ground. Even the birds that inhabit the tangle of weeds That is the roadside cover, are silent. One listens, But hears no roar of the forest. One is alone. One will be taken. One will be taken. There is no utterance, there is no conversation, But one is uneasy all the same.... There is a thin blue mist, A darkness rising like smoke, And within that darkness A possession of the heart. One will be taken.... It was here, and it will be here again- Here, under this sky empty and full of light.

Figure F.1: Four uncurated completions from a context suggesting the model compose a poem in the style of Wallace Stevens with the title ‘Shadows on the Way’.

G Details of Task Phrasing and Specifications

The following figures illustrate the formatting and phrasing of all the tasks included in the paper. All data comes from the ground truth datasets in this section, and no samples from GPT-3 are included here.

Context →	Article: Informal conversation is an important part of any business relationship. Before you start a discussion, however, make sure you understand which topics are suitable and which are considered taboo in a particular culture. Latin Americans enjoy sharing information about their local history, art and customs. You may expect questions about your family, and be sure to show pictures of your children. You may feel free to ask similar questions of your Latin American friends. The French think of conversation as an art form, and they enjoy the value of lively discussions as well as disagreements. For them, arguments can be interesting and they can cover pretty much or any topic ---- as long as they occur in a respectful and intelligent manner. In the United States, business people like to discuss a wide range of topics, including opinions about work, family, hobbies, and politics. In Japan, China, and Korea, however, people are much more private. They do not share much about their thoughts, feelings, or emotions because they feel that doing so might take away from the harmonious business relationship they're trying to build. Middle Easterners are also private about their personal lives and family matters. It is considered rude, for example, to ask a businessman from Saudi Arabia about his wife or children. As a general rule, it's best not to talk about politics or religion with your business friends. This can get you into trouble, even in the United States, where people hold different religious views. In addition, discussing one's salary is usually considered unsuitable. Sports is typically a friendly subject in most parts of the world, although be careful not to criticize national sport. Instead, be friendly and praise your host's team. Q: What shouldn't you do when talking about sports with colleagues from another country? A: Criticizing the sports of your colleagues' country. Q: Which is typically a friendly topic in most places according to the author? A: Sports. Q: Why are people from Asia more private in their conversation with others? A: They don't want to have their good relationship with others harmed by informal conversation. Q: The author considers politics and religion . . A:
Correct Answer →	taboo
Incorrect Answer →	cheerful topics
Incorrect Answer →	rude topics
Incorrect Answer →	topics that can never be talked about

Figure G.1: Formatted dataset example for RACE-h. When predicting, we normalize by the unconditional probability of each answer as described in 2.

Context →	anli 2: anli 2: The Gold Coast Hotel & Casino is a hotel and casino located in Paradise, Nevada. This locals' casino is owned and operated by Boyd Gaming. The Gold Coast is located one mile (~ 1.6km) west of the Las Vegas Strip on West Flamingo Road. It is located across the street from the Palms Casino Resort and the Rio All Suite Hotel and Casino. Question: The Gold Coast is a budget-friendly casino. True, False, or Neither?
Correct Answer →	Neither
Incorrect Answer →	True
Incorrect Answer →	False

Figure G.2: Formatted dataset example for ANLI R2

Context →	Article: Mrs. Smith is an unusual teacher. Once she told each student to bring along a few potatoes in plastic bag. On each potato the students had to write a name of a person that they hated And the next day, every child brought some potatoes. Some had two potatoes;some three;some up to five. Mrs. Smith then told the children to carry the bags everywhere they went, even to the toilet, for two weeks. As day after day passed, the children started to complain about the awful smell of the rotten potatoes. Those children who brought five potatoes began to feel the weight trouble of the bags. After two weeks, the children were happy to hear that the game was finally ended. Mrs. Smith asked,"How did you feel while carrying the potatoes for two weeks?" The children started complaining about the trouble loudly. Then Mrs. Smith told them why she asked them to play the game. She said,"This is exactly the situation when you carry your hatred for somebody inside your heart. The terrible smell of the hatred will pollute your heart and you will carry something unnecessary with you all the time. If you cannot stand the smell of the rotten potatoes for just two weeks, can you imagine how heavy it would be to have the hatred in your heart for your lifetime? So throw away any hatred from your heart, and you'll be really happy." Q: Which of the following is True according to the passage? A: If a kid hated four people,he or she had to carry four potatoes. Q: We can learn from the passage that we should _ . A: throw away the hatred inside Q: The children complained about _ besides the weight trouble. A: the smell Q: Mrs.Smith asked her students to write _ on the potatoes. A:
Correct Answer →	names
Incorrect Answer →	numbers
Incorrect Answer →	time
Incorrect Answer →	places

Figure G.3: Formatted dataset example for RACE-m. When predicting, we normalize by the unconditional probability of each answer as described in 2.

Context →	How to apply sealant to wood.
Correct Answer →	Using a brush, brush on sealant onto wood until it is fully saturated with the sealant.
Incorrect Answer →	Using a brush, drip on sealant onto wood until it is fully saturated with the sealant.

Figure G.4: Formatted dataset example for PIQA

Context →	My body cast a shadow over the grass because
Correct Answer →	the sun was rising.
Incorrect Answer →	the grass was cut.

Figure G.5: Formatted dataset example for COPA

Context →	(CNN) Yuval Rabin, whose father, Yitzhak Rabin, was assassinated while serving as Prime Minister of Israel, criticized Donald Trump for appealing to "Second Amendment people" in a speech and warned that the words that politicians use can incite violence and undermine democracy. "Trump's words are an incitement to the type of political violence that touched me personally," Rabin wrote in USA Today. He said that Trump's appeal to "Second Amendment people" to stop Hillary Clinton -- comments that were criticized as a call for violence against Clinton, something Trump denied -- "were a new level of ugliness in an ugly campaign season." - The son of a former Israeli Prime Minister who was assassinated wrote an op ed about the consequence of violent political rhetoric. - Warns of "parallels" between Israel of the 1990s and the U.S. today.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Donald Trump's aggressive rhetoric.
Correct Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Trump's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Hillary Clinton's aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned U.S.'s aggressive rhetoric.
Incorrect Answer →	- Referencing his father, who was shot and killed by an extremist amid political tension in Israel in 1995, Rabin condemned Yitzhak Rabin's aggressive rhetoric.

Figure G.6: Formatted dataset example for ReCoRD. We consider the context above to be a single "problem" because this is how the task is presented in the ReCoRD dataset and scored in the ReCoRD evaluation script.

Context →	anli 1: anli 1: Fulton James MacGregor MSP is a Scottish politician who is a Scottish National Party (SNP) Member of Scottish Parliament for the constituency of Coatbridge and Chryston. MacGregor is currently Parliamentary Liaison Officer to Shona Robison, Cabinet Secretary for Health & Sport. He also serves on the Justice and Education & Skills committees in the Scottish Parliament. Question: Fulton James MacGregor is a Scottish politician who is a Liaison officer to Shona Robison who he swears is his best friend. True, False, or Neither?
Correct Answer →	Neither
Incorrect Answer →	True
Incorrect Answer →	False

Figure G.7: Formatted dataset example for ANLI R1

Context →	Organisms require energy in order to do what?
Correct Answer →	mature and develop.
Incorrect Answer →	rest soundly.
Incorrect Answer →	absorb light.
Incorrect Answer →	take in nutrients.

Figure G.8: Formatted dataset example for OpenBookQA. When predicting, we normalize by the unconditional probability of each answer as described in 2.

Context →	Making a cake: Several cake pops are shown on a display. A woman and girl are shown making the cake pops in a kitchen. They
Correct Answer →	bake them, then frost and decorate.
Incorrect Answer →	taste them as they place them on plates.
Incorrect Answer →	put the frosting on the cake as they pan it.
Incorrect Answer →	come out and begin decorating the cake as well.

Figure G.9: Formatted dataset example for HellaSwag

Context →	anli 3: anli 3: We shut the loophole which has American workers actually subsidizing the loss of their own job. They just passed an expansion of that loophole in the last few days: \$43 billion of giveaways, including favors to the oil and gas industry and the people importing ceiling fans from China. Question: The loophole is now gone True, False, or Neither?
Correct Answer →	False
Incorrect Answer →	True
Incorrect Answer →	Neither

Figure G.10: Formatted dataset example for ANLI R3

Context →	Question: George wants to warm his hands quickly by rubbing them. Which skin surface will produce the most heat? Answer:
Correct Answer →	dry palms
Incorrect Answer →	wet palms
Incorrect Answer →	palms covered with oil
Incorrect Answer →	palms covered with lotion

Figure G.11: Formatted dataset example for ARC (Challenge). When predicting, we normalize by the unconditional probability of each answer as described in 2.

Context →	lull is to trust as
Correct Answer →	cajole is to compliance
Incorrect Answer →	balk is to fortitude
Incorrect Answer →	betray is to loyalty
Incorrect Answer →	hinder is to destination
Incorrect Answer →	soothe is to passion

Figure G.12: Formatted dataset example for SAT Analogies

Correct Context →	Grace was happy to trade me her sweater for my jacket. She thinks the sweater
Incorrect Context →	Grace was happy to trade me her sweater for my jacket. She thinks the jacket
Target Completion →	looks dowdy on her.

Figure G.13: Formatted dataset example for Winograd. The ‘partial’ evaluation method we use compares the probability of the completion given a correct and incorrect context.

Correct Context →	Johnny likes fruits more than vegetables in his new keto diet because the fruits
Incorrect Context →	Johnny likes fruits more than vegetables in his new keto diet because the vegetables
Target Completion →	are saccharine.

Figure G.14: Formatted dataset example for Winogrande. The ‘partial’ evaluation method we use compares the probability of the completion given a correct and incorrect context.

Context →	READING COMPREHENSION ANSWER KEY While this process moved along, diplomacy continued its rounds. Direct pressure on the Taliban had proved unsuccessful. As one NSC staff note put it, "Under the Taliban, Afghanistan is not so much a state sponsor of terrorism as it is a state sponsored by terrorists." In early 2000, the United States began a high-level effort to persuade Pakistan to use its influence over the Taliban. In January 2000, Assistant Secretary of State Karl Inderfurth and the State Department's counterterrorism coordinator, Michael Sheehan, met with General Musharraf in Islamabad, dangling before him the possibility of a presidential visit in March as a reward for Pakistani cooperation. Such a visit was coveted by Musharraf, partly as a sign of his government's legitimacy. He told the two envoys that he would meet with Mullah Omar and press him on Bin Laden. They left, however, reporting to Washington that Pakistan was unlikely in fact to do anything," given what it sees as the benefits of Taliban control of Afghanistan." President Clinton was scheduled to travel to India. The State Department felt that he should not visit India without also visiting Pakistan. The Secret Service and the CIA, however, warned in the strongest terms that visiting Pakistan would risk the President's life. Counterterrorism officials also argued that Pakistan had not done enough to merit a presidential visit. But President Clinton insisted on including Pakistan in the itinerary for his trip to South Asia. His one-day stopover on March 25, 2000, was the first time a U.S. president had been there since 1969. At his meeting with Musharraf and others, President Clinton concentrated on tensions between Pakistan and India and the dangers of nuclear proliferation, but also discussed Bin Laden. President Clinton told us that when he pulled Musharraf aside for a brief, one-on-one meeting, he pleaded with the general for help regarding Bin Laden." I offered him the moon when I went to see him, in terms of better relations with the United States, if he'd help us get Bin Laden and deal with another issue or two." The U.S. effort continued. Who did The State Department feel should visit both India and Pakistan?
Correct Answer →	- [False] Bin Laden
Incorrect Answer →	- [True] Bin Laden

Figure G.15: Formatted dataset example for MultiRC. There are three levels within MultiRC: (1) the passage, (2) the questions, and (3) the answers. During evaluation, accuracy is determined at the per-question level, with a question being considered correct if and only if all the answers within the question are labeled correctly. For this reason, we use K to refer to the number of **questions** shown within the context.

Context →	Question: Which factor will most likely cause a person to develop a fever? Answer:
Correct Answer →	a bacterial population in the bloodstream
Incorrect Answer →	a leg muscle relaxing after exercise
Incorrect Answer →	several viral particles on the skin
Incorrect Answer →	carbohydrates being digested in the stomach

Figure G.16: Formatted dataset example for ARC (Easy). When predicting, we normalize by the unconditional probability of each answer as described in 2.

Context →	Bob went to the gas station to fill up his car. His tank was completely empty and so was his wallet. The cashier offered to pay for his gas if he came back later to pay. Bob felt grateful as he drove home.
Correct Answer →	Bob believed that there were good people in the world.
Incorrect Answer →	Bob contemplated how unfriendly the world was.

Figure G.17: Formatted dataset example for StoryCloze

Context →	Helsinki is the capital and largest city of Finland. It is in the region of Uusimaa, in southern Finland, on the shore of the Gulf of Finland. Helsinki has a population of , an urban population of , and a metropolitan population of over 1.4 million, making it the most populous municipality and urban area in Finland. Helsinki is some north of Tallinn, Estonia, east of Stockholm, Sweden, and west of Saint Petersburg, Russia. Helsinki has close historical connections with these three cities. The Helsinki metropolitan area includes the urban core of Helsinki, Espoo, Vantaa, Kauniainen, and surrounding commuter towns. It is the world's northernmost metro area of over one million people, and the city is the northernmost capital of an EU member state. The Helsinki metropolitan area is the third largest metropolitan area in the Nordic countries after Stockholm and Copenhagen, and the City of Helsinki is the third largest after Stockholm and Oslo. Helsinki is Finland's major political, educational, financial, cultural, and research center as well as one of northern Europe's major cities. Approximately 75% of foreign companies that operate in Finland have settled in the Helsinki region. The nearby municipality of Vantaa is the location of Helsinki Airport, with frequent service to various destinations in Europe and Asia. Q: what is the most populous municipality in Finland? A: Helsinki Q: how many people live there? A: 1.4 million in the metropolitan area Q: what percent of the foreign companies that operate in Finland are in Helsinki? A: 75% Q: what towns are a part of the metropolitan area? A:
Target Completion →	Helsinki, Espoo, Vantaa, Kauniainen, and surrounding commuter towns

Figure G.18: Formatted dataset example for CoQA

Context →	Please unscramble the letters into a word, and write that word: asinoc =
Target Completion →	casino

Figure G.19: Formatted dataset example for Cycled Letters

Context →	Passage: Saint Jean de Brébeuf was a French Jesuit missionary who travelled to New France in 1625. There he worked primarily with the Huron for the rest of his life, except for a few years in France from 1629 to 1633. He learned their language and culture, writing extensively about each to aid other missionaries. In 1649, Brébeuf and another missionary were captured when an Iroquois raid took over a Huron village . Together with Huron captives, the missionaries were ritually tortured and killed on March 16, 1649. Brébeuf was beatified in 1925 and among eight Jesuit missionaries canonized as saints in the Roman Catholic Church in 1930. Question: How many years did Saint Jean de Brébeuf stay in New France before he went back to France for a few years? Answer:
Target Completion →	4

Figure G.20: Formatted dataset example for DROP

Context →	Fill in blank: She held the torch in front of her. She caught her breath. "Chris? There's a step." "What?" "A step. Cut in the rock. About fifty feet ahead." She moved faster. They both moved faster. "In fact," she said, raising the torch higher, "there's more than a ----. ->
Target Completion →	step

Figure G.21: Formatted dataset example for LAMBADA

Context →	Please unscramble the letters into a word, and write that word: skicts =
Target Completion →	sticks

Figure G.22: Formatted dataset example for Anagrams 1 (A1)

Context →	Please unscramble the letters into a word, and write that word: volwskagen =
Target Completion →	volkswagen

Figure G.23: Formatted dataset example for Anagrams 2

Context →	Q: Who played tess on touched by an angel? A:
Target Completion →	Delloreese Patricia Early (July 6, 1931 { November 19, 2017), known professionally as Della Reese

Figure G.24: Formatted dataset example for Natural Questions

Context →	TITLE: William Perry (American football) - Professional career PARAGRAPH: In 1985, he was selected in the first round of the 1985 NFL Draft by the Chicago Bears; he had been hand-picked by coach Mike Ditka. However, defensive coordinator Buddy Ryan, who had a highly acrimonious relationship with Ditka, called Perry a "wasted draft-pick". Perry soon became a pawn in the political power struggle between Ditka and Ryan. Perry's "Refrigerator" nickname followed him into the NFL and he quickly became a favorite of the Chicago Bears fans. Teammates called him "Biscuit," as in "one biscuit shy of 350 pounds." While Ryan refused to play Perry, Ditka decided to use Perry as a fullback when the team was near the opponents' goal line or in fourth and short situations, either as a ball carrier or a lead blocker for star running back Walter Payton. Ditka stated the inspiration for using Perry as a fullback came to him during five-yard sprint exercises. During his rookie season, Perry rushed for two touchdowns and caught a pass for one. Perry even had the opportunity to run the ball during Super Bowl XX, as a nod to his popularity and contributions to the team's success. The first time he got the ball, he was tackled for a one-yard loss while attempting to throw his first NFL pass on a halfback option play. The second time he got the ball, he scored a touchdown (running over Patriots linebacker Larry McGrew in the process). About halfway through his rookie season, Ryan finally began to play Perry, who soon proved that he was a capable defensive lineman. His Super Bowl ring size is the largest of any professional football player in the history of the event. His ring size is 25, while the ring size for the average adult male is between 10 and 12. Perry went on to play for ten years in the NFL, retiring after the 1994 season. In his ten years as a pro, he regularly struggled with his weight, which hampered his performance at times. He played in 138 games, recording 29.5 sacks and five fumble recoveries, which he returned for a total of 71 yards. In his offensive career he ran five yards for two touchdowns, and had one reception for another touchdown. Perry later attempted a comeback, playing an unremarkable 1996 season with the London Monarchs of the World League of American Football (later NFL Europa). Q: what team did he play for? A:
Target Completion →	the Chicago Bears

Figure G.25: Formatted dataset example for QuAC

Context →	Please unscramble the letters into a word, and write that word: r e!c.i p r o.c a/l =
Target Completion →	reciprocal

Figure G.26: Formatted dataset example for Symbol Insertion

Context →	Please unscramble the letters into a word, and write that word: taefed =
Target Completion →	defeat

Figure G.27: Formatted dataset example for Reversed Words

Context →	Title: The_Blitz
	Background: From the German point of view, March 1941 saw an improvement. The Luftwaffe flew 4,000 sorties that month, including 12 major and three heavy attacks. The electronic war intensified but the Luftwaffe flew major inland missions only on moonlit nights. Ports were easier to find and made better targets. To confuse the British, radio silence was observed until the bombs fell. X- and Y-Gerät beams were placed over false targets and switched only at the last minute. Rapid frequency changes were introduced for X-Gerät, whose wider band of frequencies and greater tactical flexibility ensured it remained effective at a time when British selective jamming was degrading the effectiveness of Y-Gerät.
	Q: How many sorties were flown in March 1941?
	A: 4,000
	Q: When did the Luftwaffe fly inland missions?
	A:
Target Completion →	only on moonlit nights

Figure G.28: Formatted dataset example for SQuADv2

Context →	Normal force -- In a simple case such as an object resting upon a table, the normal force on the object is equal but in opposite direction to the gravitational force applied on the object (or the weight of the object), that is, $N = m g$ ($\displaystyle N=mg$), where m is mass, and g is the gravitational field strength (about 9.81 m/s on Earth). The normal force here represents the force applied by the table against the object that prevents it from sinking through the table and requires that the table is sturdy enough to deliver this normal force without breaking. However, it is easy to assume that the normal force and weight are action-reaction force pairs (a common mistake). In this case, the normal force and weight need to be equal in magnitude to explain why there is no upward acceleration of the object. For example, a ball that bounces upwards accelerates upwards because the normal force acting on the ball is larger in magnitude than the weight of the ball. question: is the normal force equal to the force of gravity? answer:
Target Completion →	yes

Figure G.29: Formatted dataset example for BoolQ

Context →	The trend toward lower rents may seem surprising given that some communities in New York are bemoaning the loss of favorite local businesses to high rents. But, despite the recent softening, for many of these retailers there's still been too big a jump from the rental rates of the late 1970s, when their leases were signed. Certainly, the recent drop in prices doesn't mean Manhattan comes cheap. question: Manhattan comes cheap. true, false, or neither? answer:
Target Completion →	false

Figure G.30: Formatted dataset example for CB

Context →	The bet, which won him dinner for four, was regarding the existence and mass of the top quark, an elementary particle discovered in 1995. question: The Top Quark is the last of six flavors of quarks predicted by the standard model theory of particle physics. True or False? answer:
Target Completion →	False

Figure G.31: Formatted dataset example for RTE

Context →	An outfitter provided everything needed for the safari. Before his first walking holiday, he went to a specialist outfitter to buy some boots. question: Is the word 'outfitter' used in the same way in the two sentences above? answer:
Target Completion →	no

Figure G.32: Formatted dataset example for WiC

Context →	Final Exam with Answer Key Instructions: Please carefully read the following passages. For each passage, you must identify which noun the pronoun marked in *bold* refers to. ===== Passage: Mr. Moncrieff visited Chester's luxurious New York apartment, thinking that it belonged to his son Edward. The result was that Mr. Moncrieff has decided to cancel Edward's allowance on the ground that he no longer requires *his* financial support. Question: In the passage above, what does the pronoun " *his* " refer to? Answer:
Target Completion →	mr. moncrieff

Figure G.33: Formatted dataset example for WSC

Context →	Q: 'Nude Descending A Staircase' is perhaps the most famous painting by which 20th century artist? A:
Target Completion →	MARCEL DUCHAMP
Target Completion →	r mutt
Target Completion →	duchamp
Target Completion →	marcel duchamp
Target Completion →	R.Mutt
Target Completion →	Marcel duChamp
Target Completion →	Henri-Robert-Marcel Duchamp
Target Completion →	Marcel du Champ
Target Completion →	henri robert marcel duchamp
Target Completion →	Duchampian
Target Completion →	Duchamp
Target Completion →	duchampian
Target Completion →	marcel du champ
Target Completion →	Marcel Duchamp
Target Completion →	MARCEL DUCHAMP

Figure G.34: Formatted dataset example for TriviaQA. TriviaQA allows for multiple valid completions.

Context →	Q: What school did burne hogarth establish?
	A:
Target Completion →	School of Visual Arts

Figure G.35: Formatted dataset example for WebQA

Context →	Keinesfalls dürfen diese für den kommerziellen Gebrauch verwendet werden. =
Target Completion →	In no case may they be used for commercial purposes.

Figure G.36: Formatted dataset example for De→En. This is the format for one- and few-shot learning, for this and other language tasks, the format for zero-shot learning is “Q: What is the {language} translation of {sentence} A: {translation}.”

Context →	In no case may they be used for commercial purposes. =
Target Completion →	Keinesfalls dürfen diese für den kommerziellen Gebrauch verwendet werden.

Figure G.37: Formatted dataset example for En→De

Context →	Analysis of instar distributions of larval I. verticalis collected from a series of ponds also indicated that males were in more advanced instars than females. =
Target Completion →	L’analyse de la distribution de fréquence des stades larvaires d’I. verticalis dans une série d’étangs a également démontré que les larves mâles étaient à des stades plus avancés que les larves femelles.

Figure G.38: Formatted dataset example for En→Fr

Context →	L’analyse de la distribution de fréquence des stades larvaires d’I. verticalis dans une série d’étangs a également démontré que les larves mâles étaient à des stades plus avancés que les larves femelles. =
Target Completion →	Analysis of instar distributions of larval I. verticalis collected from a series of ponds also indicated that males were in more advanced instars than females.

Figure G.39: Formatted dataset example for Fr→En

Context →	The truth is that you want, at any price, and against the wishes of the peoples of Europe, to continue the negotiations for Turkey’s accession to the European Union, despite Turkey’s continuing refusal to recognise Cyprus and despite the fact that the democratic reforms are at a standstill. =
Target Completion →	Adevărul este că vă doriți, cu orice preț și împotriva dorinței europenilor, să continuați negocierile de aderare a Turciei la Uniunea Europeană, în ciuda refuzului continuu al Turciei de a recunoaște Ciprul și în ciuda faptului că reformele democratice au ajuns într-un punct mort.

Figure G.40: Formatted dataset example for En→Ro

Context →	Adevărul este că vă doriți, cu orice preț și împotriva dorinței europenilor, să continuați negocierile de aderare a Turciei la Uniunea Europeană, în ciuda refuzului continuu al Turciei de a recunoaște Ciprul și în ciuda faptului că reformele democratice au ajuns într-un punct mort. =
Target Completion →	The truth is that you want, at any price, and against the wishes of the peoples of Europe, to continue the negotiations for Turkey's accession to the European Union, despite Turkey's continuing refusal to recognise Cyprus and despite the fact that the democratic reforms are at a standstill.

Figure G.41: Formatted dataset example for Ro→En

Context →	Q: What is $(2 * 4) * 6$? A:
Target Completion →	48

Figure G.42: Formatted dataset example for Arithmetic 1DC

Context →	Q: What is 17 minus 14? A:
Target Completion →	3

Figure G.43: Formatted dataset example for Arithmetic 2D-

Context →	Q: What is 98 plus 45? A:
Target Completion →	143

Figure G.44: Formatted dataset example for Arithmetic 2D+

Context →	Q: What is 95 times 45? A:
Target Completion →	4275

Figure G.45: Formatted dataset example for Arithmetic 2Dx

Context →	Q: What is 509 minus 488? A:
Target Completion →	21

Figure G.46: Formatted dataset example for Arithmetic 3D-

Context →	Q: What is 556 plus 497? A:
Target Completion →	1053

Figure G.47: Formatted dataset example for Arithmetic 3D+

Context →	Q: What is 6209 minus 3365? A:
Target Completion →	2844

Figure G.48: Formatted dataset example for Arithmetic 4D-

Context →	Q: What is 9923 plus 617? A:
Target Completion →	10540

Figure G.49: Formatted dataset example for Arithmetic 4D+

Context →	Q: What is 40649 minus 78746? A:
Target Completion →	-38097

Figure G.50: Formatted dataset example for Arithmetic 5D−

Context →	Q: What is 65360 plus 16204? A:
Target Completion →	81564

Figure G.51: Formatted dataset example for Arithmetic 5D+

H Results on All Tasks for All Model Sizes

					Zero-Shot								One-Shot								Few-Shot								175B
Name	Metric	Split	Fine-tune																										(test server)
			SOTA	K	Small	Med	Large	XL	2.7B	6.7B	13B	175B	Small	Med	Large	XL	2.7B	6.7B	13B	175B	Small	Med	Large	XL	2.7B	6.7B	13B	175B	
HellaSwag	acc	dev	85.6	20	33.7	43.6	51.0	54.7	62.8	67.4	70.9	78.9	33.0	42.9	50.5	53.5	61.9	66.5	70.0	78.1	33.5	43.1	51.3	54.9	62.9	67.3	71.3	79.3	
LAMBADA	acc	test	68.0	15	42.7	54.3	60.4	63.6	67.1	70.3	72.5	76.2	22.0	47.1	52.6	58.3	61.1	65.4	69.0	72.5	22.0	40.4	63.2	57.0	78.1	79.1	81.3	86.4	
LAMBADA	ppl	test	8.63	15	18.6	9.09	6.53	5.44	4.60	4.00	3.56	3.00	165.0	11.6	8.29	6.46	5.53	4.61	4.06	3.35	165.0	27.6	6.63	7.45	2.89	2.56	2.56	1.92	
StoryCloze	acc	test	91.8	70	63.3	68.5	72.4	73.4	77.2	77.7	79.5	83.2	62.3	68.7	72.3	74.2	77.3	78.7	79.7	84.7	62.3	70.2	73.9	76.1	80.2	81.2	83.0	87.7	
NQs	acc	test	44.5	64	0.64	1.75	2.71	4.40	6.01	5.79	7.84	14.6	1.19	3.07	4.79	5.43	8.73	9.78	13.7	23.0	1.72	4.46	7.89	9.72	13.2	17.0	21.0	29.9	71.2
TriviaQA	acc	dev	68.0	64	4.15	7.61	14.0	19.7	31.3	38.7	41.8	64.3	4.19	12.9	20.5	26.5	35.9	44.4	51.3	68.0	6.96	16.3	26.5	32.1	42.3	51.6	57.5	71.2	
WebQs	acc	test	45.5	64	1.77	3.20	4.33	4.63	7.92	7.73	8.22	14.4	2.56	6.20	8.51	9.15	14.5	15.1	19.0	25.3	5.46	12.6	15.9	19.6	24.8	27.7	33.5	41.5	
Ro→En 16	BLEU-mb	test	39.9	64	2.08	2.71	3.09	3.15	16.3	8.34	20.2	19.9	0.55	15.4	23.0	26.3	30.6	33.2	35.6	38.6	1.25	20.7	25.8	29.2	33.1	34.8	37.0	39.5	
Ro→En 16	BLEU-sb	test		64	2.39	3.08	3.49	3.56	16.8	8.75	20.8	20.9	0.65	15.9	23.6	26.8	31.3	34.2	36.7	40.0	1.40	21.3	26.6	30.1	34.3	36.2	38.4	41.3	
En→Ro 16	BLEU-mb	test	38.5	64	2.14	2.65	2.53	2.50	3.46	4.24	5.32	14.1	0.35	3.30	7.89	8.72	13.2	15.1	17.3	20.6	1.25	5.90	9.33	10.7	14.3	16.3	18.0	21.0	
En→Ro 16	BLEU-sb	test		64	2.61	3.11	3.07	3.09	4.26	5.31	6.43	18.0	0.55	3.90	9.15	10.3	15.7	18.2	20.8	24.9	1.64	7.40	10.9	12.9	17.2	19.6	21.8	25.8	
Fr→En 14	BLEU-mb	test	35.0	64	1.81	2.53	3.47	3.13	20.6	15.1	21.8	21.2	1.28	15.9	23.7	26.3	29.0	30.5	30.2	33.7	4.98	25.5	28.5	31.1	33.7	34.9	36.6	39.2	
Fr→En 14	BLEU-sb	test		64	2.29	2.99	3.90	3.60	21.2	15.5	22.4	21.9	1.50	16.3	24.4	27.0	30.0	31.6	31.4	35.6	5.30	26.2	29.5	32.2	35.1	36.4	38.3	41.4	
En→Fr 14	BLEU-mb	test	45.6	64	1.74	2.16	2.73	2.15	15.1	8.82	12.0	25.2	0.49	8.00	14.8	15.9	20.3	23.3	24.9	28.3	4.08	14.5	19.3	21.5	24.9	27.3	29.5	32.6	
En→Fr 14	BLEU-sb	test	45.9	64	2.44	2.75	3.54	2.82	19.3	11.4	15.3	31.3	0.81	10.0	18.2	19.3	24.7	28.3	30.1	34.1	5.31	18.0	23.6	26.1	30.3	33.3	35.5	39.9	
De→En 16	BLEU-mb	test	40.2	64	2.06	2.87	3.41	3.63	21.5	17.3	23.0	27.2	0.83	16.2	22.5	24.7	28.2	30.7	33.0	30.4	3.25	22.7	26.2	29.2	32.7	34.8	37.3	40.6	
De→En 16	BLEU-sb	test		64	2.39	3.27	3.85	4.04	22.5	18.2	24.4	28.6	0.93	17.1	23.4	25.8	29.2	31.9	34.5	32.1	3.60	23.8	27.5	30.5	34.1	36.5	39.1	43.0	
En→De 16	BLEU-mb	test	41.2	64	1.70	2.27	2.31	2.43	12.9	8.66	10.4	24.6	0.50	7.00	12.9	13.1	18.3	20.9	22.5	26.2	3.42	12.3	15.4	17.1	20.9	23.0	26.6	29.7	
En→De 16	BLEU-sb	test	41.2	64	2.09	2.65	2.75	2.92	13.7	9.36	11.0	25.3	0.54	7.40	13.4	13.4	18.8	21.7	23.3	27.3	3.78	12.9	16.1	17.7	21.7	24.1	27.7	30.9	
Winograd	acc	test	93.8	7	66.3	72.9	74.7	76.9	82.4	85.7	87.9	88.3	63.4	68.5	72.9	76.9	82.4	84.6	86.1	89.7	63.4	67.4	73.6	76.9	84.3	85.4	82.4	88.6	82.8
Winogrande	acc	dev	84.6	50	52.0	52.1	57.4	58.7	62.3	64.5	67.9	70.2	51.3	53.0	58.3	59.1	61.7	65.8	66.9	73.2	51.3	52.6	57.5	59.1	62.6	67.4	70.0	77.7	
PIQA	acc	dev	77.1	50	64.6	70.2	72.9	75.1	75.6	78.0	78.5	81.0	64.3	69.3	71.8	74.4	74.3	76.3	77.8	80.5	64.3	69.4	72.0	74.3	75.4	77.8	79.9	82.3	
ARC (Challenge)	acc	test	78.5	50	26.6	29.5	31.8	35.5	38.0	41.4	43.7	51.4	25.5	30.2	31.6	36.4	38.4	41.5	43.1	53.2	25.5	28.4	32.3	36.7	39.5	43.7	44.8	51.5	
ARC (Easy)	acc	test	92.0	50	43.6	46.5	53.0	53.8	58.2	60.2	63.8	68.8	42.7	48.2	54.6	55.9	60.3	62.6	66.8	71.2	42.7	51.0	58.1	59.1	62.1	65.8	69.1	70.1	82.8
OpenBookQA	acc	test	87.2	100	35.6	43.2	45.2	46.8	53.0	50.4	55.6	57.6	37.0	39.8	46.2	46.4	53.4	53.0	55.8	58.8	37.0	43.6	48.0	50.6	55.6	55.2	60.8	65.4	
Quac	f1	dev	74.4	5	21.2	26.8	31.0	30.1	34.7	36.1	38.4	41.5	21.1	26.9	31.9	32.3	37.4	39.0	40.6	43.4	21.6	27.6	32.9	34.2	38.2	39.9	40.9	44.3	
RACE-h	acc	test	90.0	10	35.2	37.9	40.1	40.9	42.4	44.1	44.6	45.5	34.3	37.7	40.0	42.0	43.8	44.3	44.6	45.9	34.3	37.0	40.4	41.4	42.3	44.7	45.1	46.8	
RACE-m	acc	test	93.1	10	42.1	47.2	52.1	52.3	54.7	54.4	56.7	58.4	42.3	47.3	51.7	55.2	56.1	54.7	56.9	57.4	42.3	47.0	52.7	53.0	55.6	55.4	58.1	58.1	
SQuADv2	em	dev	90.7	16	22.6	32.8	33.9	43.1	43.6	45.4	49.0	52.6	25.1	37.5	37.9	47.9	47.9	51.1	56.0	60.1	27.5	40.5	39.2	53.5	50.0	56.6	62.6	64.9	
SQuADv2	f1	dev	93.0	16	28.3	40.2	41.4	50.3	51.0	52.7	56.3	59.5	30.1	43.6	44.1	54.0	54.1	57.1	61.8	65.4	32.1	45.5	44.9	58.7	55.9	62.1	67.7	69.8	
CoQA	f1	dev	90.7	5	34.5	55.0	61.8	65.3	71.1	72.8	76.3	81.5	30.6	52.1	61.6	66.1	71.8	75.1	77.9	84.0	31.1	52.0	62.7	66.8	73.2	77.3	79.9	85.0	
DROP	f1	dev	89.1	20	9.40	13.6	14.4	16.4	19.7	17.0	24.0	23.6	11.7	18.1	20.9	23.0	26.4	27.3	29.2	34.3	12.9	18.7	24.0	25.6	29.7	29.7	32.3	36.5	
BoolQ	acc	dev	91.0	32	49.7	60.3	58.9	62.4	67.1	65.4	66.2	60.5	52.6	61.7	60.4	63.7	68.4	68.7	69.0	76.7	43.1	60.6	62.0	64.1	70.3	70.0	70.2	77.5	76.4
CB	acc	dev	96.9	32	0.00	32.1	8.93	19.6	19.6	28.6	19.6	46.4	55.4	53.6	53.6	48.2	57.1	33.9	55.4	64.3	42.9	58.9	53.6	69.6	67.9	60.7	66.1	82.1	75.6
CB	f1	dev	93.9	32	0.00	29.3	11.4	17.4	22.4	25.1	20.3	42.8	60.1	39.8	45.6	37.5	45.7	28.5	44.6	52.5	26.1	40.4	32.6	48.3	45.7	44.6	46.0	57.2	52.0
Copa	acc	dev	94.8	32	66.0	68.0	73.0	77.0	76.0	80.0	84.0	91.0	62.0	64.0	66.0	74.0	76.0	82.0	86.0	87.0	67.0	64.0	72.0	77.0	83.0	83.0	86.0	92.0	92.0
RTE	acc	dev	92.5	32	47.7	49.8	48.4	56.0	46.6	55.2	62.8	63.5	53.1	47.3	49.5	49.5	54.9	54.9	56.3	70.4	52.3	48.4	46.9	50.9	56.3	49.5	60.6	72.9	69.0
WiC	acc	dev	76.1	32	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.0	50.3	50.3	49.2	49.4	50.3	50.0	48.6	49.8	55.0	53.0	53.0	51.6	53.1	51.1	55.3	49.4
WSC	acc	dev	93.8	32	59.6	56.7	65.4	61.5	66.3	60.6	64.4	65.4	58.7	58.7	60.6	62.5	66.3	60.6	66.3	69.2	58.7	60.6	54.8	49.0	62.5	67.3	75.0	75.0	80.1
MultiRC	acc	dev	62.3	32	4.72	9.65	12.3	13.6	14.3	18.4	24.2	27.6	4.72	9.65	12.3	13.6	14.3	18.4	24.2	27.6	6.09	11.8	16.8	20.8	24.7	23.8	25.0	32.5	30.5
MultiRC	f1a	dev	88.2	32	57.0	59.7	60.4	59.9	60.0	64.5	71.4	72.9	57.0	59.7	60.4	59.9	60.0	64.5	71.4	72.9	45.0	55.9	64.2	65.4	69.5	66.4			

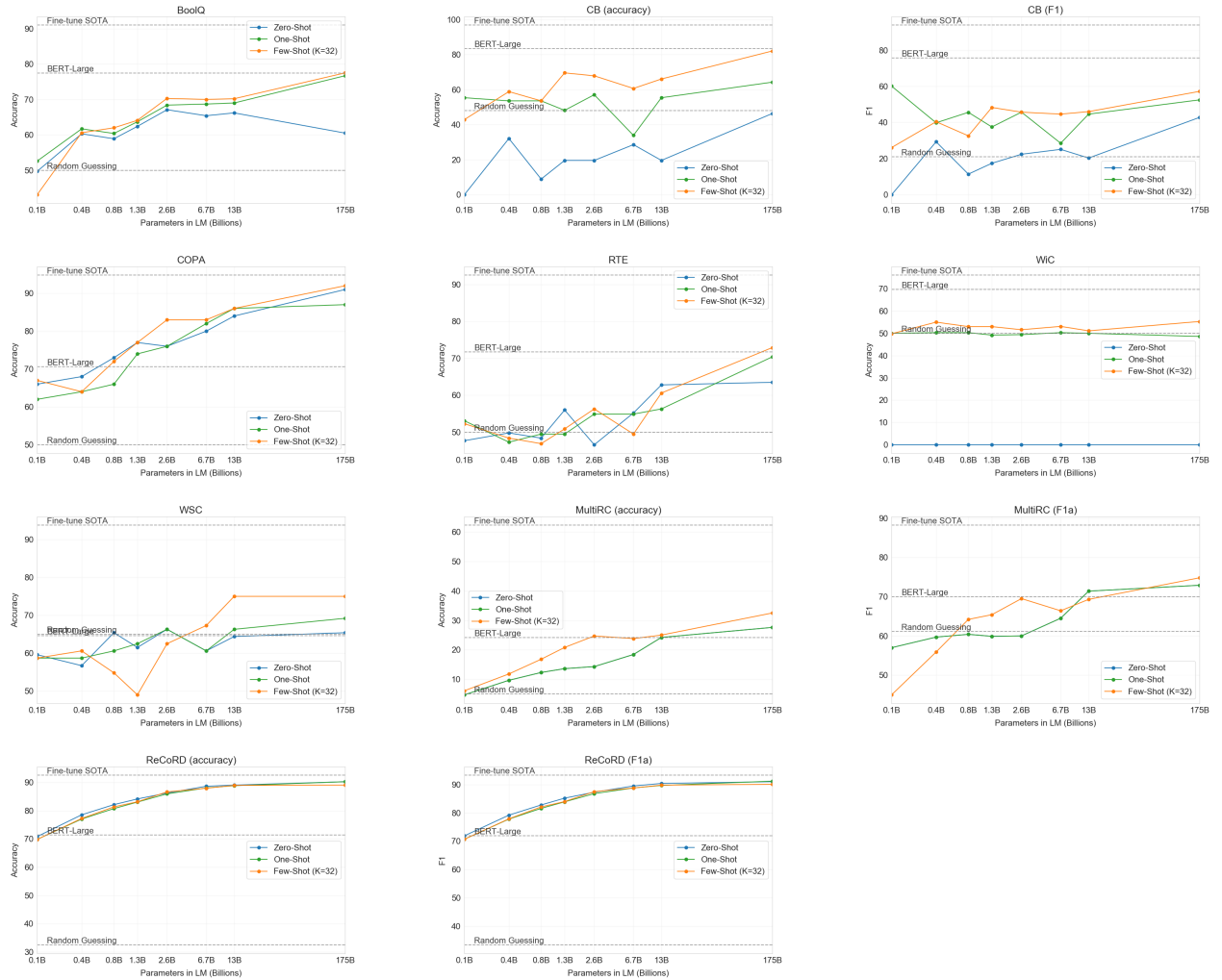


Figure H.1: All results for all SuperGLUE tasks.

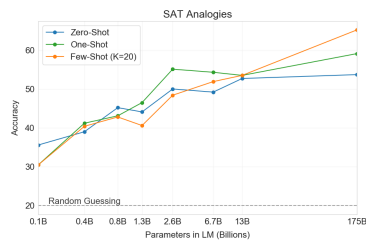


Figure H.2: Results for SAT task.

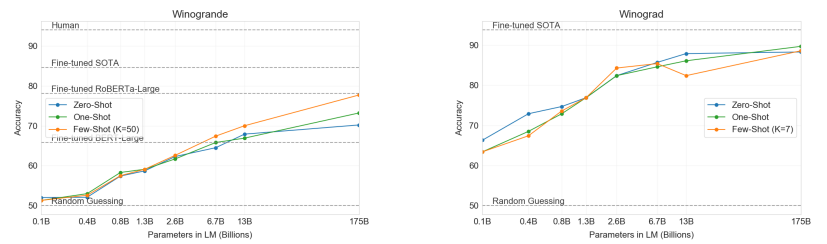


Figure H.3: All results for all Winograd tasks.

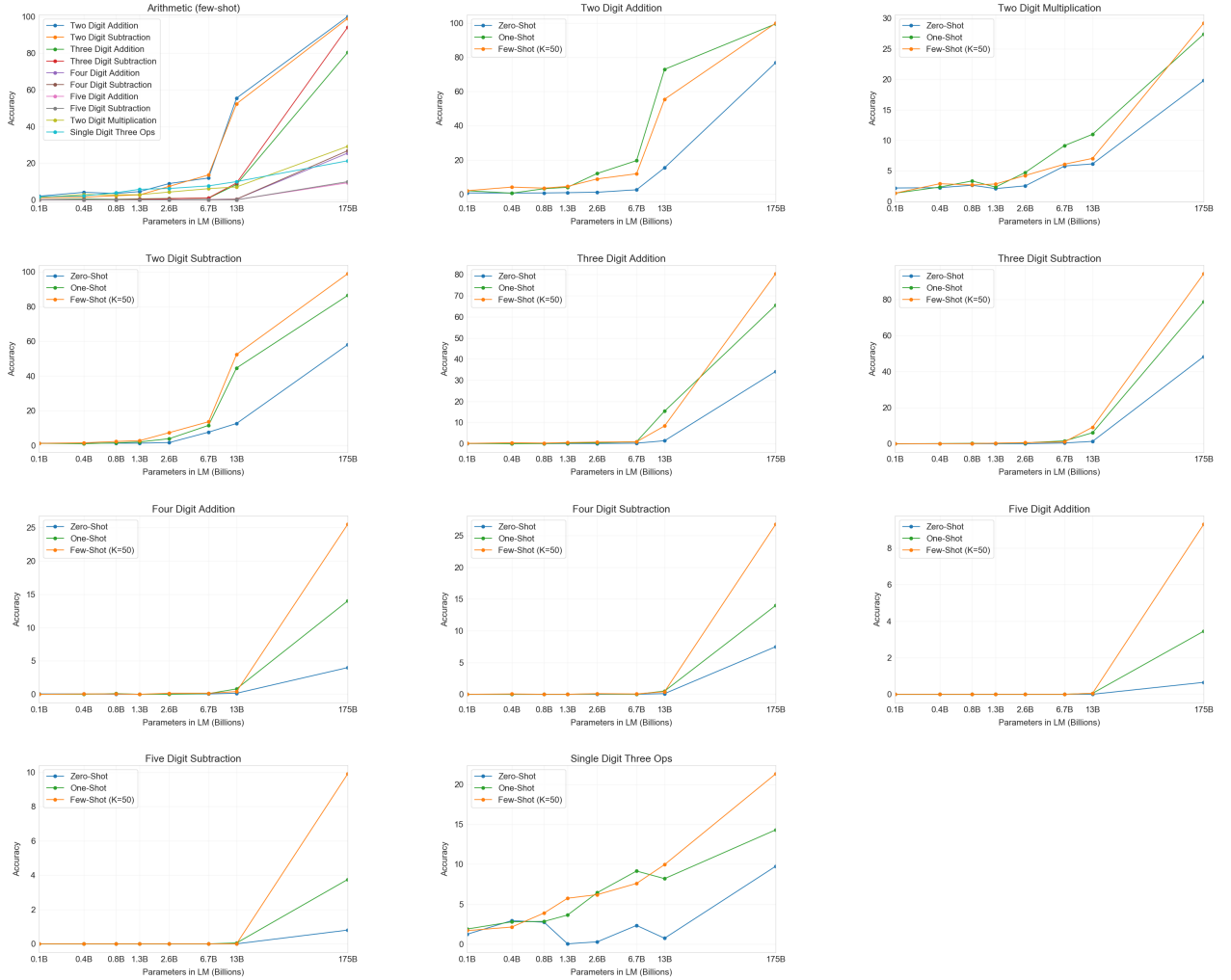


Figure H.4: All results for all Arithmetic tasks.

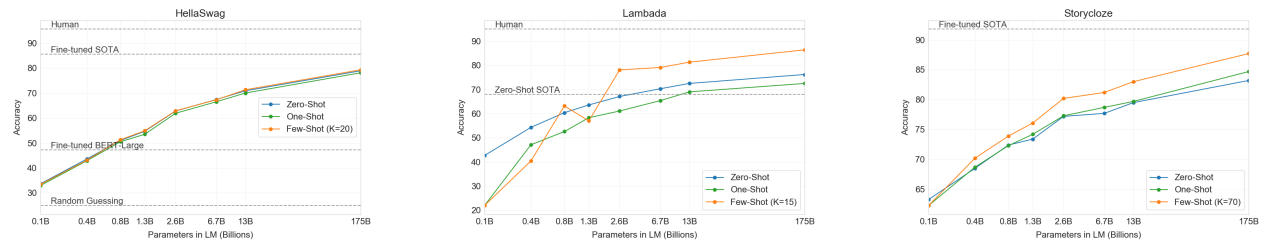


Figure H.5: All results for all Cloze and Completion tasks.

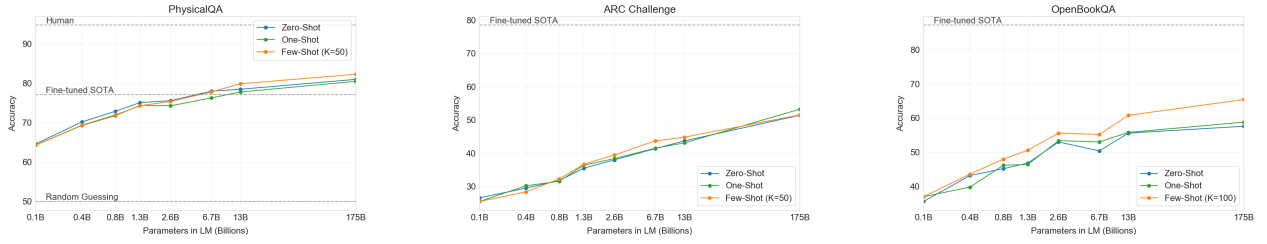


Figure H.6: All results for all Common Sense Reasoning tasks.

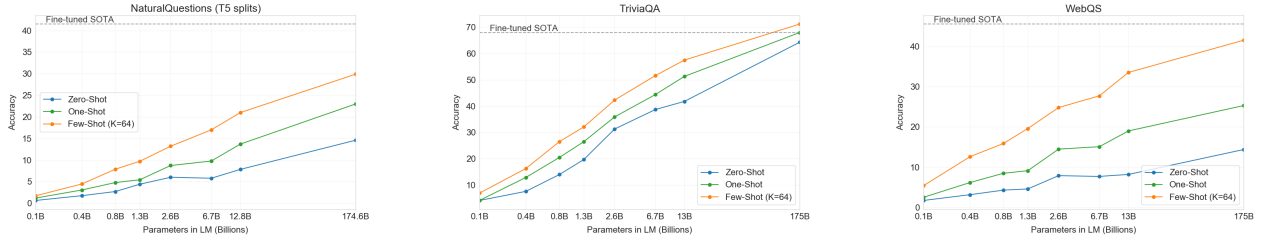


Figure H.7: All results for all QA tasks.

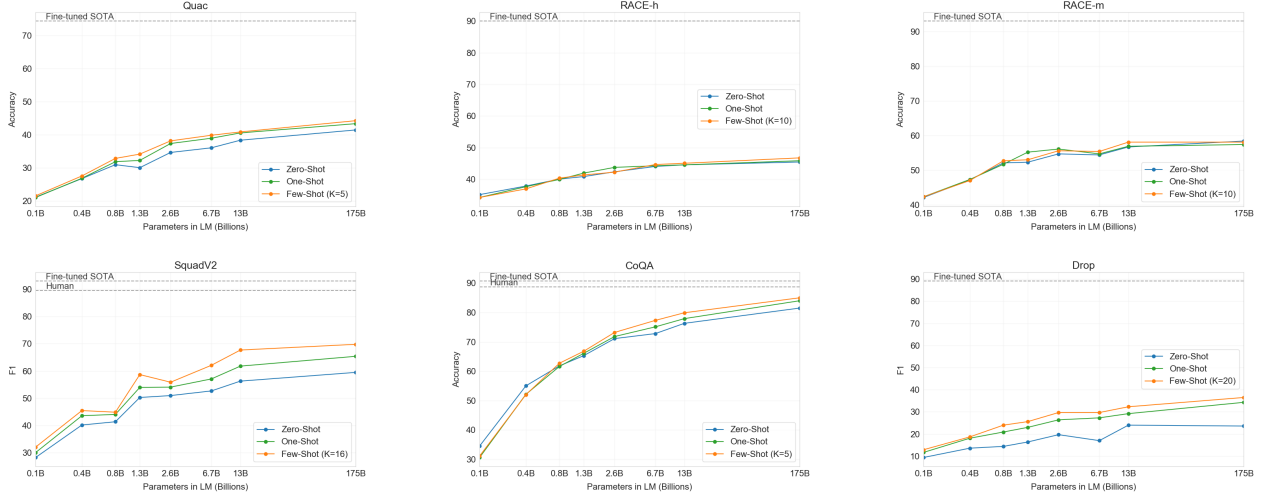


Figure H.8: All results for all Reading Comprehension tasks.

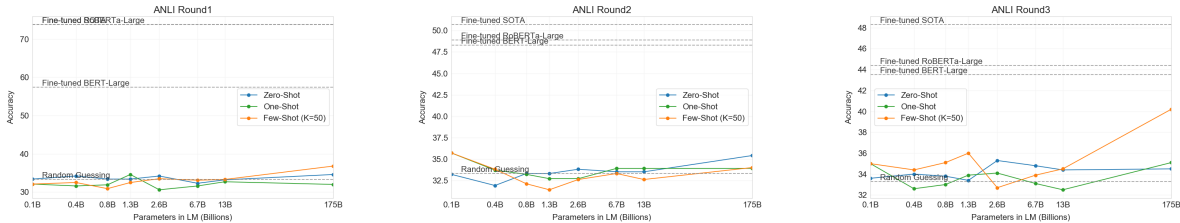


Figure H.9: All results for all ANLI rounds.

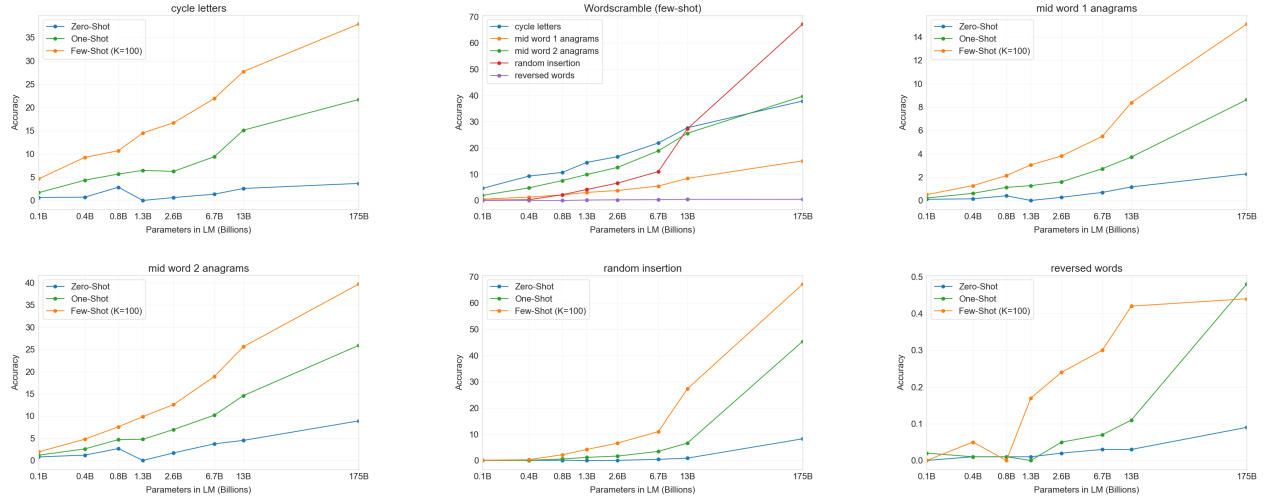


Figure H.10: All results for all Scramble tasks.

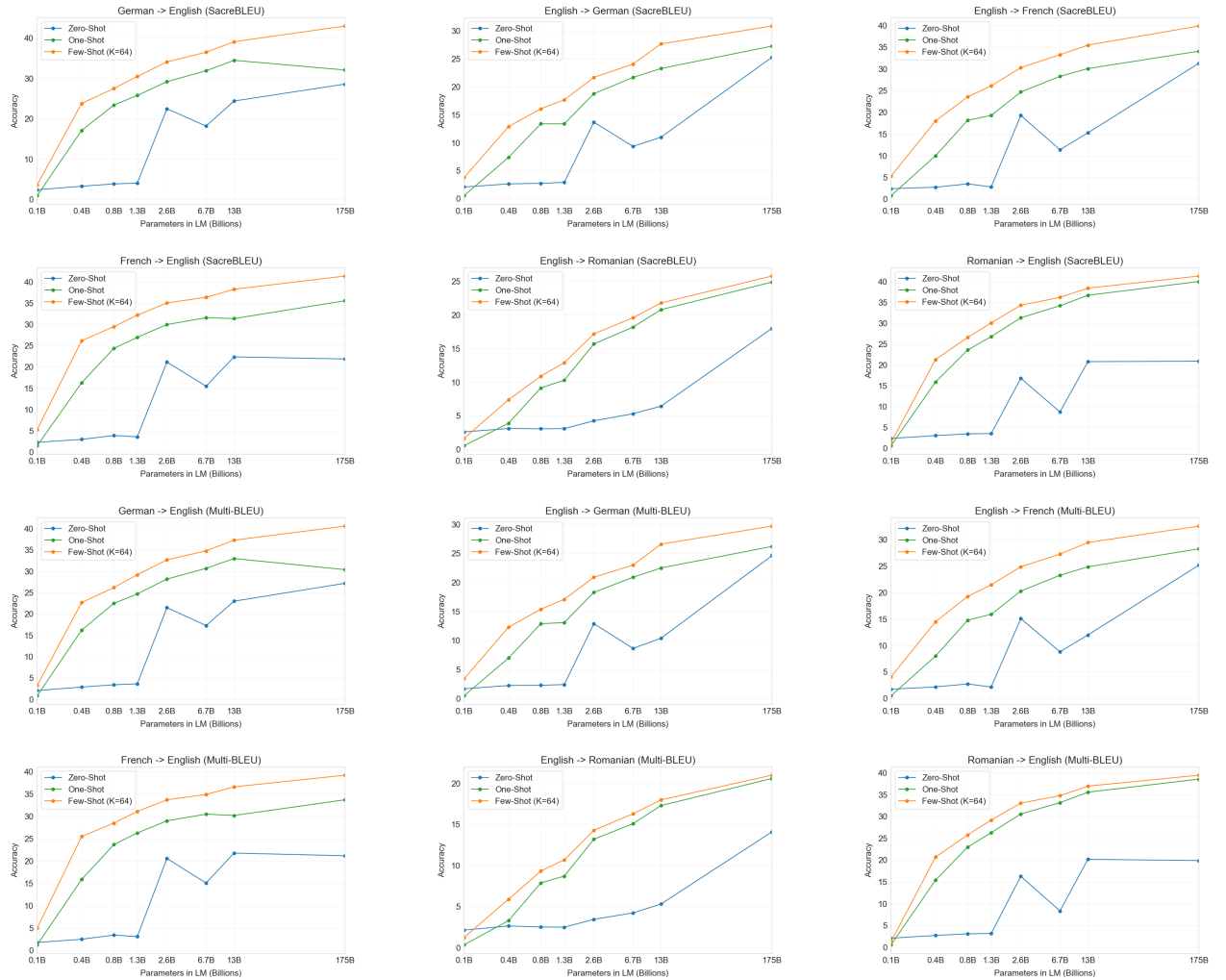


Figure H.11: All results for all Translation tasks.

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